

HW 3



3. (a).

We have loss function.

$$L(f(x), y) = L_0(f(x)=1, y=0) + L_1(f(x)=0, y=1)$$

objective function: $\min \mathbb{E}(L(f(x), y))$

$$\begin{aligned}\Rightarrow \arg \min_{f(x)} \mathbb{E}(L(f(x), y)) &= \arg \min \mathbb{E}[L_0(f(x)=1, y=0) + L_1(f(x)=0, y=1)] \\ &= \arg \min \mathbb{E}[L_0(f(x)=1, y=0) + L_1(f(x)=0, y=1) | x=x] \\ &= \arg \min \mathbb{E}[\sum L_0(f(x)=1) \cdot P(y=0|x) \\ &\quad + \sum L_1(f(x)=0) \cdot P(y=1|x)]\end{aligned}$$

$$\Rightarrow f(x) = \begin{cases} 1 & , L_0 P(y=0|x=x) \leq L_1 P(y=1|x=x) \\ 0 & , \text{otherwise} \end{cases}$$

(b) $\because L_1 > L_0$

$\therefore$ The cost of guessing $y=1$ when it is actually $y=0$ is less than the cost of guessing $y=0$ when it is actually $y=1$, which means MORE penalty is given when we guess $y=0$ that actually it is 1.

$\therefore P(L_1) < \frac{1}{2} < P(L_0)$ to minimize total loss.

(c) By part (a), we know in this case $f(x) = 1$

$$\therefore L_0 P(y=1|x=x) \geq L_1 P(y=0|x=x)$$

$$\text{let } P(y=1|x=x) = b, \text{ then } P(y=0|x=x) = 1-b$$

$$\therefore \frac{b}{1-b} \geq \frac{L_1}{L_0}$$

Take $\log$ on both sides, $\log\left(\frac{b}{1-b}\right) \geq \log\left(\frac{L_1}{L_0}\right)$, where $\log\left(\frac{b}{1-b}\right) = x^T \hat{\beta}$

$\therefore$ The rule becomes:

$$\hat{f}(x) = \begin{cases} 1, & x^T \hat{\beta} \geq \log\left(\frac{L_1}{L_0}\right) \\ 0, & \text{otherwise} \end{cases}$$

$\therefore$ when $L_0 > L_1$, the boundary is $\log\left(\frac{L_1}{L_0}\right)$.